Yugo Iwamoto

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Overview

Human–Computer Interaction (HCI) researcher passionate about supporting real-world needs, specializing in human-centered AI, accessibility, and learning technologies. I design and evaluate human-centered AI accessibility workflows (e.g., assisting non-experts in generating and verifying alt text for complex scientific figures), assistive communication with d/Deaf and hard-of-hearing individuals (e.g., Kiosk Multimodal communications Tool), and learning technologies to support educators. Across projects, I apply iterative prototyping and mixed-methods research with individuals with disabilities and accessibility professionals to produce deployable systems and evidence-based design guidance.

Research Interests: Human–Computer Interaction (HCI); Human-AI Interaction and Human-Centered AI, Accessibility, Assistive Technology, and Inclusive Learning Technologies, Co-Design / Participatory Methods, Mixed-Methods Evaluation, Generative AI in Accessibility

Education

M.S. in Human-Computer Interaction, Rochester Institute of Technology (RIT), NY, US

Aug 2023 – Aug 2025 | CGPA: 3.78

Capstone: Bridge Communications: Kiosk Multimodal Communications tool for hearing and d/DHH users.

Relevant Coursework: Research Methods, Usability Testing, Wearable and IoT, Designing Information, Web Development I & II, Foundation of HCI

B.S. in Communications & Recreation (Recreation and Tourism), State University of New York (SUNY) at Brockport, NY, US

Jan 2018 – Dec 2021 | CGPA: 3.37

Relevant Coursework: Advertising Communications, Tourism Marketing, Foundation of Tourism Marketing, Interpersonal Communications, Recreation Management, Conflict Management

Research Experience

NSF AWARE-AI Research Trainee – AWARE-AI NRT, RIT, Aug 2024 – Present

- Human Computer Interaction Track under Dr. Jamison Heard and Dr. Garreth Tigwell
- Designed and evaluated GenAI-assisted workflows for scientific-figure image descriptions with disability service professionals (mixed-methods: rubric scoring + interviews).

- Analyzed mixed methods data, both qualitative and quantitative, and collaborated with disability service professionals.
- Accepted Work in Progress/Poster paper at ASSETS 2025: "Exploring Generative AI to Support Disability Service Professionals in Writing Image Descriptions for HCI Science Figures."
- Applied participatory design principles in evaluating generative AI-supported tools with accessibility professionals.

Graduate Research Assistant – School of Information, RIT, Aug 2024 – May 2025

- Assisted Associate Professor Catherine Beaton
- Designed and evaluated inclusive digital platforms for faculty with disabilities and launched a website for the project.
- Developed and prepared generative AI for K-12 underfunded school workshops, targeting teachers to enrich their course materials.
- Led studies on GenAI in K-12 and multilingual STEM education.

Capstone Project – Kiosk real-time speech-to-text tool for inclusive conversations between hearing and DHH users, Aug 2024 – Present | Rochester Institute of Technology, NY

- Committee Chair: Dr. Tae Oh and Committee Member: Dr. Michael McQuaid
- Developed a real-time speech-to-text prototype using an API to support conversations between Deaf/Hard-of-Hearing (DHH) and hearing individuals. Prototype is here
- Conducted user-centered design research and usability testing with DHH and hearing participants and analyzed both qualitative and quantitative data (NASA TLX, interviews).
- Focused on inclusive communication, assistive technology, and accessibility innovation in everyday environments.

Selected Publications & Presentations

1. **Y. Iwamoto**, M. Raees, T. Oh, and M. McQuaid. 2026. Bridge Communications: Kiosk Multimodal Communication Tool for d/Deaf and Hard of Hearing and Hearing. In Proceedings of the 28th International Conference on Human-Computer Interaction (HCII' 26), Springer
2. M. Raees, **Y. Iwamoto*,** K. Pagangelis, J. Heard, and G Tigwell. 2026. How University Disability Services Professionals Write Image Description for HCI Figures Using Generative AI. *arXiv preprint arXiv:2602.08937*
3. **Y. Iwamoto**, M. Raees*, J. Heard, and G. Tigwell. 2025. Exploring Generative AI to Support Disability Service Professionals in Writing Image Descriptions for HCI Science Figures. In Proceedings of the 28th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '25). ACM, New York, NY, USA. https://doi.org/10.1145/3663547.3759763
4. **Y. Iwamoto**, C. Beaton, and A. Roberts. 2025. Digital Literacy and AI in Underfunded Schools. In the IEEE STEM Education Conference (Workshop), Princeton University. IEEE.

5. **Y. Iwamoto**, A. Roberts, R. Mahajan, C. Beaton. 2025. Importance of Diverse Role Models in Education to Support Equal Education Opportunity and Inclusivity. *2025 IEEE Integrated STEM Education Conference (ISEC)*, 1–6. https://doi.org/10.1109/isec64801.2025.11147317
6. **Y. Iwamoto** and M. Raees. 2025. A11y: Improving Alt-Text for Science Figures with GenAI. In the Graduate Showcase (Poster & Oral Presentation), Rochester Institute of Technology.
7. C. Beaton, E. Weeden, A. Roberts, and **Y. Iwamoto**. 2024. Being Included: Institute Benefits in Supporting Inclusivity for Disabled Faculty in Higher Education. In ICERI2024 Proceedings, 5648–5654.

*Equal Contribution

Technical & UX Skills

Programming: Python, Java, HTML, CSS, JavaScript, PHP, MySQL, C++, JSON
Research: Research-through-design, thematic analysis, mixed-methods research, statistical analysis, accessibility audits, participatory design, user study, literature review, qualitative research,
UX & Design: Figma, Wireframing, Usability Testing, Adobe CC, Responsive Design
Tools: WordPress, Blackboard, Arduino, JMP, Microsoft 365, Google Suite, Render

Professional Experience

Graduate Assistant – *Disability Services Office, Rochester Institute of Technology*
Jan 2024 – May 2025 | Rochester, NY

- Provided accessibility support and assistive technology solutions for students with disabilities.
- Ensured equitable access by coordinating academic accommodations across various disciplines.
- Supported exam proctoring, technical troubleshooting, and content conversion for screen reader compatibility.

Communications & Student Experience Coordinator – *RIT School of Information*
Jan 2022 – Dec 2022 | Rochester, NY

- Led digital content strategy and social media; supervised student ambassadors.
- Managed web projects, content creation, and user-centered site enhancements.
- Support research publication promotion

Communications Intern – *Gates Chili Central School District*
Jan 2021 – Jun 2021 | Rochester, NY

- Created multilingual and accessible digital content for community engagement.
- Updated and maintained web presence with a focus on inclusive design.
- Supported communication across cultural and linguistic backgrounds.

Receptionist – *Woodcliff Hotel and Spa*

Aug 2021 – Dec 2021 | Fairport, NY

- Handled guest services, events, and hospitality logistics.
- Supported banquet team for weddings and special events as a server, people mover driver, and assisting servers
- Supported revenue management from a guest service standpoint.

Purser, High-Speed Rail Attendant – *Japan Rail Central Passengers*
May 2018 – Aug 2020 | Osaka, Japan

- Delivered bilingual customer service and coordinated on-board operations.
- Assisting passengers with special needs and improving the experience of the trip.
- Ensured efficient, safe, and punctual operation of bullet train services.

Leadership & Community Engagement

1. **International Orientation Leader** – SUNY Brockport, 2018–2020
2. **President of the International Student Organization** – SUNY Brockport, 2018-2021
3. **Global Ambassador for Study Abroad Programs** – SUNY Brockport, 2019–2020

Achievements, Honors & Awards

1. **Graduate Research Assistant Scholarship** (2024–2025)
2. **Dean's List Honors** – SUNY Brockport (2019–2021)
3. **International Transfer Student Merit-Based Scholarship** (2018-2020)
4. **GCCSD Pride Award** – Gates Chili Central School District (2021)
5. **Employee of the Month**- Woodcliff Hotel and Spa (2021)

Funding

1. **Graduate School Research & Creative Fundings**: – RIT; $500 (2025)

Language

1. English– Native
2. Japanese– Native
3. German-Basic (currently learning)

Academic/ Research Reference

1. **Dr. Jamison Heard** (jrheee@rit.edu) – Assistant Professor at RIT, Supervisor for NSF AWARE AI Traineeship
2. **Dr. Catherine Beaton** (ciiics@rit.edu) – Associate Professor at RIT, Supervisor for Graduate Research Assistant

3. **Dr. Tae Oh** (thoics@rit.edu) – Professor at RIT, Committee Chair for MS Capstone Project
4. **Dr. Michael McQuaid** (micmcquaid@gmail.com) – Associate Professor at UT Austin, MS Capstone Committee Member
5. **Mrs. Betty Hillman** (echics@rit.edu) – Assistant Director at RIT, Previous Supervisor for Coordinator of Communications and Student Experience